

Calcolo vettoriale e tensoriale

$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$

$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a})$

$\varepsilon_{ijk}\varepsilon_{ipq} = \delta_{jp}\delta_{kq} - \delta_{jq}\delta_{kp}$

$(\mathbf{a} \times \mathbf{b})_i = \varepsilon_{ijk}\mathbf{a}_j\mathbf{b}_k, \quad \mathbf{a} \cdot \mathbf{b} = \mathbf{a}_i\mathbf{b}_i = \delta_{ij}\mathbf{a}_i\mathbf{b}_j$

Coordinate naturali e polari

$\mathbf{v} = \dot{s}\hat{\boldsymbol{\tau}} \quad \mathbf{a} = \ddot{s}\hat{\boldsymbol{\tau}} + \frac{\dot{s}^2}{\rho}\hat{\boldsymbol{\rho}} \quad \hat{\boldsymbol{b}} = \hat{\boldsymbol{\tau}} \times \hat{\boldsymbol{\rho}}$

$\frac{d\hat{\boldsymbol{\tau}}}{ds} = \frac{1}{\rho}\hat{\boldsymbol{\rho}}, \quad \frac{d\hat{\boldsymbol{\rho}}}{ds} = -\frac{1}{\rho}\hat{\boldsymbol{\tau}} + \frac{1}{\sigma}\hat{\boldsymbol{\phi}}, \quad \frac{d\hat{\boldsymbol{\phi}}}{ds} = -\frac{1}{\sigma}\hat{\boldsymbol{\rho}}$

$\mathbf{a} = \hat{\boldsymbol{\tau}}(\ddot{r} - r\dot{\theta}^2) + \hat{\boldsymbol{\theta}}(r\ddot{\theta} + 2\dot{r}\dot{\theta})$

Equazioni cardinali

$M\mathbf{a}_{CM} = \frac{d\mathbf{P}_{TOT}}{dt} = \mathbf{F}^{ext}, \quad \frac{d\mathbf{L}_O}{dt} = \mathbf{M}_O^{ext} - \mathbf{v}_O \times \mathbf{P}_{TOT}$

Equazione di Binet e orbita planetaria

$\mu = \frac{Mm}{M+m}, \quad V_{eff} = U + \frac{\ell^2}{2mr^2}$

$\frac{\ell^2 u^2}{m} \left[\frac{d^2 u}{d\theta^2} + u \right] = -f\left(\frac{1}{u}\right)$

$r(\theta) = \frac{p}{1 + e \cos(\theta)}, \quad p = \frac{\ell^2}{\alpha m}, \quad e = \sqrt{1 + \frac{2E\ell^2}{m\alpha^2}}$
 $\frac{(1 - e^2)^2}{p^2} \left(x + \frac{pe}{1 - e^2} \right)^2 + y^2 \left(\frac{1 - e^2}{p^2} \right) = 1 \quad E = -\frac{\alpha}{2a}$

Vettore di Lenz

$\mathbf{N} = \frac{\dot{\mathbf{r}} \times \mathbf{L}}{\alpha} - \hat{\mathbf{r}}, \quad |\mathbf{N}| = \sqrt{1 + \frac{2E\ell^2}{m\alpha^2}}$

Sistemi rotanti e non inerziali

$\boldsymbol{\omega} = \hat{\mathbf{i}}' \left(-\hat{\mathbf{j}}' \cdot \frac{d\hat{\mathbf{k}}'}{dt} \right) + \hat{\mathbf{j}}' \left(-\hat{\mathbf{k}}' \cdot \frac{d\hat{\mathbf{i}}'}{dt} \right) + \hat{\mathbf{k}}' \left(-\hat{\mathbf{i}}' \cdot \frac{d\hat{\mathbf{j}}'}{dt} \right)$

A.I.R. parallelo a $\boldsymbol{\omega}$ e passante per $\mathbf{r}'_Q - \frac{1}{\omega^2}(\mathbf{v}_Q \times \boldsymbol{\omega})$

$m\mathbf{a}' = \mathbf{F} - m[\mathbf{a}_{O'} + \dot{\boldsymbol{\omega}} \times \mathbf{r}' + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}') + 2\boldsymbol{\omega} \times \mathbf{v}']$

$\left(\frac{d\mathbf{b}}{dt} \right)_{spazio} = \left(\frac{d\mathbf{b}}{dt} \right)_{corpo} + \boldsymbol{\omega} \times \mathbf{b}$

$\mathbf{v}_P = \mathbf{v}_{O'} + \mathbf{v}'_P + \boldsymbol{\omega} \times \mathbf{r}'_P$

Corpo rigido

$\mathcal{I}^{(P)} = \sum_{A=1}^N m_A \begin{pmatrix} y_A^2 + z_A^2 & -x_A y_A & -x_A z_A \\ -x_A y_A & x_A^2 + z_A^2 & -y_A z_A \\ -x_A z_A & -y_A z_A & x_A^2 + y_A^2 \end{pmatrix}$

Asse fisso $\hat{\mathbf{n}}$: $I_n^{(P)} = \sum_{A=1}^N m_A \left[\mathbf{r}_A^2 - (\mathbf{r}_A \cdot \hat{\mathbf{n}})^2 \right]$

Eulero: $M_x^{ext} = I_x \dot{\omega}_x + (I_z - I_y) \omega_z \omega_y$

Teorema di Steiner: $I_n^{(P)} = Md^2 + I_n^{(CM)}$

Piccole oscillazioni

$q_i = q_i^{(0)} + \eta_i, \quad \boldsymbol{\eta} = (\eta_1, \dots, \eta_n)^\top$

$V_{jk} = \frac{\partial^2 U}{\partial q_j \partial q_k} \Big|_{q^{(0)}}, \quad T_{jk} = \sum_{\ell} m_{\ell} \left(\frac{\partial \mathbf{r}_{\ell}}{\partial q_j} \cdot \frac{\partial \mathbf{r}_{\ell}}{\partial q_k} \right) \Big|_{q^{(0)}}$

$T_{osc} = \frac{1}{2} \dot{\boldsymbol{\eta}}^\top T \dot{\boldsymbol{\eta}} \quad U_{osc} \approx \frac{1}{2} \boldsymbol{\eta}^\top V \boldsymbol{\eta} \quad \det(V - \omega^2 T) = 0$

$(V_{ij} - \omega_k^2 T_{ij}) A_{jk} = 0 \quad A_{ij} \zeta_j = \eta_i$

Equazioni del moto di Hamilton

$\mathcal{H}(q_i, p_i, t) = \sum_i \dot{q}_i p_i - \mathcal{L}(q_i, \dot{q}_i, t)$

$\dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i} \quad \dot{p}_i = -\frac{\partial \mathcal{H}}{\partial q_i} \quad \frac{d\mathcal{H}}{dt} = \frac{\partial \mathcal{H}}{\partial t} = -\frac{\partial \mathcal{L}}{\partial t}$

Parentesi di Poisson

$\{F, G\} = \sum_i \left(\frac{\partial F}{\partial q_i} \frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial G}{\partial q_i} \right)$

$\{q_i, q_j\} = 0, \quad \{p_i, p_j\} = 0, \quad \{q_i, p_j\} = \delta_{ij}$

$\frac{df}{dt} = \{f, \mathcal{H}\} + \frac{\partial f}{\partial t}$

Trasformazioni canoniche

$\mathcal{K} = \mathcal{H} + \frac{\partial F_i}{\partial t} \quad \forall i$

$F_1(q, Q, t) \quad p_i = \frac{\partial F_1}{\partial q_i} \quad P_i = -\frac{\partial F_1}{\partial Q_i}$

$F_2(q, P, t) \quad p_i = \frac{\partial F_2}{\partial q_i} \quad Q_i = \frac{\partial F_2}{\partial P_i}$

$F_3(p, Q, t) \quad q_i = -\frac{\partial F_3}{\partial p_i} \quad P_i = -\frac{\partial F_3}{\partial Q_i}$

$F_4(p, P, t) \quad q_i = -\frac{\partial F_4}{\partial p_i} \quad Q_i = \frac{\partial F_4}{\partial P_i}$